# DAKSHESH GAUTAM

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## PROFESSIONAL SUMMARY

Data and Machine Learning enthusiast with hands-on experience in Python, SQL, exploratory data analysis, feature engineering, predictive modeling, and model evaluation. Built projects spanning fraud detection, deep learning, and neural-network-based estimation, including real-time ML inference and model integration with simulation systems. Interested in applying data-driven methods to solve practical business, analytical, and engineering problems across Data Science and Machine Learning roles.

## TECHNICAL SKILLS

**Programming:** Python, SQL

**Machine Learning:** Scikit-learn, XGBoost, Gradient Boosting, Random Forest, SVM, KNN, MLP, Feature Engineering

**Deep Learning:** TensorFlow, Keras, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN)

**Data Analysis:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Exploratory Data Analysis

**Tools:** MATLAB, Simulink, Flask, Docker, Render, Advance Excel, Git, GitHub, Linux

## PROJECTS

**Transaction Risk Detector (Fraud Detection)** *Python, XGBoost, Flask, Docker, Render*

- Developed an end-to-end fraud detection system with a **3-stage inference pipeline** covering input handling, Pandas preprocessing, and XGBoost prediction.
- Processed **14 transaction and customer attributes** including transaction amount, demographics, merchant details, geolocation, and time features for fraud-risk prediction.
- Addressed severe class imbalance in a **2-class fraud classification** problem using `scale_pos_weight` and assessed performance with **PR-AUC** for rare-fraud detection.
- Deployed a Dockerized real-time inference service on Render, returning **2 outputs**: fraud probability and risk classification, with logging and exception handling.

**ANN-Based Speed & Position Estimation for BLDC Motor** *MATLAB, Simulink, ANN*

- Generated and collected BLDC motor operating data from a **MATLAB/Simulink model** for developing a neural-network-based speed and position estimation system.
- Preprocessed the simulated motor data and trained an **Artificial Neural Network (ANN)** to estimate motor speed and rotor position.
- Integrated the trained ANN into the **BLDC Simulink model** to perform speed and position estimation during motor operation.
- Validated the integrated estimator by comparing **estimated speed and position** with the corresponding reference motor outputs.

**Handwritten Digit Recognition (Computer Vision)** *Python, TensorFlow, Keras, scikit-learn*

- Implemented a Convolutional Neural Network (CNN) for handwritten digit classification using the MNIST dataset.
- **Compared 7 classification models** for MNIST recognition: CNN, Random Forest, SVM, Decision Tree, KNN, DBN, and MLP.
- Reported results across **10,000 test samples**; the CNN achieved **98.42% test accuracy** with approximately **98.0% precision, recall, and F1-score**.
- Trained the CNN for **10 epochs** and analyzed training and validation accuracy and loss curves to assess convergence and generalization.

## EDUCATION

**Netaji Subhas University of Technology (NSUT), New Delhi** *Expected 2027*
B.Tech in Electrical Engineering

**DAV Schools Network, New Delhi** *2023*
Class XII – CBSE

**S.L.S. D.A.V. Public School, New Delhi** *2021*
Class X – CBSE